

Internship Report

on

Software Engineering Intern PayU

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Bachelor of Technology in Computer Engineering

by

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CERTIFICATE

This is to certified that Internship Report entitled **Software Engineering Intern** submitted by **Satkar Garje** of B.Tech **Computer** for the work completed at **PayU, Pune** is approved for the submission. It is also certified that, to the best of my knowledge, the report represents actual work carried out by the student at the industry, company or firm.

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I am extremely great full to my department staff members and friends who helped me in successful completion of this internship.

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Declaration

This is to certify, that the Internship report submitted by me is an outcome of my independent and original work completed at PayU. I have duly acknowledged all the sources from which the ideas and extracts have been taken. The Internship report is free from any plagiarism and has not been submitted elsewhere for publication.

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Abstract

One semester internship has been completed at PayU in offline mode to study real-world backend development workflows, transaction processing systems, caching mechanisms, retry handling strategies, and testing practices used in enterprise applications. During this internship, the focus was on understanding how backend systems are optimized for performance, reliability, and scalability while handling complex transaction-related scenarios in production-oriented environments. The internship provided practical exposure to backend architecture, API optimization, failure handling mechanisms, transaction state management, regression testing, and requirement analysis processes followed in industry environments.

During the internship, multiple activities were carried out including implementation of caching mechanisms using Redis in Spring Boot applications, optimization of data retrieval processes, enhancement of currency conversion modules, development of retry logic for failed international transactions, transaction status handling, callback processing, refund scenario handling, local testing environment setup, and exploratory regression testing. Additional work involved requirement analysis and implementation support for merchant-specific risk analysis features. Backend development tasks were mainly implemented using PHP, Java, SQL, Redis, and Postman-based testing workflows.

The work was carried out by first understanding the existing system architecture and business flow through mentor discussions, documentation study, and code analysis. Tasks were divided into smaller modules and implemented incrementally with proper validation and testing. Code reviews, debugging, local testing, and optimization techniques were used throughout the internship to improve implementation quality and system reliability.

The internship provided valuable insights into enterprise backend development, transaction lifecycle management, system optimization, and real-world software engineering practices. Overall, the experience significantly improved technical skills, analytical thinking, debugging abilities, and understanding of industry-level development workflows.

Keywords: Backend Development, Caching, Regression Testing, Transaction Processing

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Chapter 1

Introduction

The internship completed at PayU focused on understanding real-world backend development workflows, transaction processing systems, caching mechanisms, retry handling strategies, testing practices, and enterprise application development processes. In modern digital platforms, backend systems play a major role in ensuring reliability, scalability, performance, and secure handling of transactions. Organizations handling high-volume transactions require optimized systems capable of managing failures, maintaining consistency, processing requests efficiently, and supporting smooth communication between different services. This internship provided practical exposure to such production-oriented backend systems and industry-level development practices.

The motivation for studying this topic came from the increasing importance of scalable backend systems and reliable transaction management in the software industry. Backend development is one of the core areas of modern applications, especially in financial and transaction-based platforms where system stability, performance optimization, and failure handling are highly important. Understanding concepts such as caching, retry mechanisms, transaction state management, callback handling, regression testing, and requirement analysis helps bridge the gap between theoretical learning and practical industry implementation.

During the internship, multiple backend development and testing activities were performed. Initial work involved optimization of data retrieval and currency conversion processes using Redis-based caching mechanisms in Spring Boot applications. Further tasks included implementation of retry logic for failed international transactions, transaction status handling, callback processing, refund scenario handling, and local testing environment setup. Exposure was also gained in regression testing practices and merchant-specific risk analysis workflows. The internship additionally involved code reviews, debugging, testing using Postman, requirement analysis, and understanding real-world software development workflows followed in enterprise systems.

The topics studied during the internship are highly relevant to engineering subjects such as Database Management Systems, Software Engineering, Object-Oriented Programming, Operating Systems, Computer Networks, Web Technologies, and Distributed Systems. Concepts learned in academics such as APIs, database operations, caching, transaction handling, system architecture, optimization techniques, and software testing were practically applied during the internship. This helped strengthen theoretical understanding through industry implementation and improved technical, analytical, and problem-solving skills.

Overall, the internship enhanced practical knowledge of backend engineering, enterprise workflows, transaction handling systems, testing methodologies, and scalable application design, thereby improving industry readiness and professional development.

1.1 Objectives

The object of this internship is to

- Gain knowledge of real-world backend development workflows and enterprise systems.
- Understand transaction processing, retry mechanisms, and backend optimization strategies.
- Learn about caching mechanisms, API handling, regression testing, and debugging techniques.
- Get exposed to production-oriented software development and testing practices.
- Relate theoretical knowledge of Database Management Systems, Software Engineering, and Object-Oriented Programming with industry implementation.
- Interact with mentors, developers, and team members in a professional environment.
- Develop backend development, testing, debugging, and analytical problem-solving skills.
- Design reliable and maintainable backend solutions for transaction-related systems.
- Prototype and implement optimized backend workflows and validation mechanisms.
- Empower oneself for future roles in backend development and software engineering.
- Develop professional communication, teamwork, and technical documentation skills.

1.2 Organization of the Internship Report

The internship report is organized as follows.

Chapter 1 gives the introduction to the internship. Also, it states the objectives of the internship. Chapter 2 provides the information of the industry/organization where the internship is completed. Chapter 3 presents the learning acquired during the internship. Chapter 4 provides the recommendations and suggestions/results of the work completed during the internship. The conclusion and the further scope are given in Chapter 5.

Chapter 2

Organization Overview

2.1 Company background

PayU is a leading fintech and digital payments company that provides online payment solutions, payment gateway services, and financial technology products for businesses and customers. The company operates across multiple countries and supports secure, fast, and scalable digital transactions for e-commerce platforms, merchants, and enterprises. PayU is known for its strong technology-driven ecosystem, secure payment infrastructure, and focus on innovation in digital finance. The internship was carried out at the Pune office of PayU in Hybrid mode.

This company was selected because it provides an opportunity to work on real-world scalable backend systems and production-level applications in the fintech domain. PayU's strong focus on technology, innovation, security, and software engineering practices made it an ideal environment to gain practical industry experience. The internship also aligned well with academic interests in backend development, software engineering, and scalable systems.

2.2 Services/products offered

PayU is primarily a payment aggregator that provides digital payment solutions for businesses and merchants. The company:

- Processes multiple modes of payment such as UPI, credit cards, debit cards, and net banking
- Provides APIs for merchants, businesses, and applications to integrate payment services
- Handles transaction processing, settlements, and refunds
- Ensures secure payment processing with fraud prevention mechanisms and security systems

2.3 Organizational structure

PayU follows a structured corporate and technical organizational hierarchy to manage product development, operations, security, and business processes efficiently. The organization consists of multiple teams working collaboratively in agile workflows.

The major departments and teams include:

- Product Management Team
- Software Development / Engineering Team
- Backend and Platform Teams
- Quality Assurance and Testing Team
- DevOps and Infrastructure Team
- Application Security (AppSec) Team
- Production Support and Operations Team
- Human Resources and Administration Team

2.4 Technologies used in the company

The major technologies, platforms, and tools used at PayU during the internship include:

- Java and Spring Boot for backend and enterprise application development
- MySQL for database management and query handling
- GitLab for version control, Merge Requests, and CI/CD workflows
- Jira for sprint planning and task tracking
- Microsoft Teams for communication and meetings
- Confluence for documentation and knowledge sharing

Chapter 3

Internship Details

3.1 Role and Responsibilities

Role

- Software Developer Intern

Tasks and Responsibilities

- Developing and enhancing backend functionalities for transaction-oriented enterprise applications using PHP, Java, and Spring Boot.
- Understanding existing backend workflows and analyzing transaction processing systems based on business requirements.
- Implementing caching mechanisms using Redis to improve data storage and retrieval performance.
- Working on retry handling logic, transaction status management, callback processing, and refund-related workflows.
- Performing backend testing, debugging, validation, and scenario-based analysis using Postman and local environments.
- Setting up and configuring local testing environments for end-to-end transaction testing.
- Exploring regression testing approaches and creating test scenarios for transaction validation.
- Participating in requirement analysis discussions and identifying edge cases in backend workflows.
- Supporting implementation of merchant-specific backend handling and response management features.
- Participating in code reviews, applying optimization suggestions, and maintaining clean coding practices.
- Preparing technical documentation, implementation notes, and workflow understanding for assigned tasks.
- Collaborating with mentors and team members during discussions, debugging, and development activities.

3.2 Tools and Technologies

- **Programming languages**
 - Java for backend development, API handling, and optimization tasks.
 - PHP for backend workflow implementation and transaction-related logic.
 - SQL for database operations, query execution, and transaction management.
- **Frameworks**
 - Spring Boot for backend service development and REST API handling.
 - Redis for caching and performance optimization mechanisms.
- **Database**

- MySQL/SQL-based databases for data storage and transaction handling.
- **Development and Testing Tools**
 - IntelliJ IDEA for backend development and debugging.
 - Postman for API testing and validation of transaction workflows.
 - Git for version control and code management.
 - Docker for local environment setup and testing workflows.
- **Collaboration and Workflow Tools**
 - Jira for task tracking and sprint-based workflow management.
 - Microsoft Teams for communication and technical discussions.
 - Confluence for documentation and knowledge sharing.
- **Other Technologies**
 - REST APIs for communication between backend services.
 - Local testing environments for end-to-end transaction validation and regression testing.

Chapter 4

Detailed Description of Internship Work

4.1 Task-wise / Module-wise Description

Task 1: Cache Monitoring and Retrieval API Development

- **Description of the task:**
Worked on implementing a backend API for monitoring and retrieving cached data using specific cache keys. The task focused on improving access to cached information and supporting faster backend operations.
- **Approach/method used:**
Analyzed the existing caching workflow and studied how cache entries were being stored and accessed. The implementation was divided into API handling, cache retrieval logic, validation, and response management.
- **Implementation details:**
Developed a Spring Boot-based API endpoint integrated with Redis for retrieving cache data using provided keys. Added validation checks, response handling, and tested multiple scenarios using Postman and local environments. Performed optimization and incorporated code review suggestions for better maintainability and reliability.
- **Output/result:**
Successfully implemented the cache retrieval API with efficient key-based access and improved monitoring support for cached backend data.

Task 2: Currency Conversion Storage and Retrieval Optimization

- **Description of the task:**
Enhanced the storage and retrieval process for currency conversion-related data to improve efficiency and response handling.
- **Approach/method used:**
Studied the existing data flow and redesigned the storage approach using caching mechanisms for optimized access and retrieval.
- **Implementation details:**
Integrated Redis-based storage handling with backend services and optimized retrieval logic. Performed validation and testing for multiple scenarios to ensure consistency and reliability.
- **Output/result:**
Successfully optimized currency conversion data handling with improved retrieval speed and reduced processing overhead.

Task 3: Retry Mechanism for Failed International Transactions

- **Description of the task:**
Worked on implementing retry handling logic for failed international transactions and managing different transaction scenarios.
- **Approach/method used:**
Analyzed transaction workflows, identified failure conditions, and divided the implementation into retry logic, transaction status handling, callback processing, and refund management modules.
- **Implementation details:**
Implemented backend logic using PHP and SQL for retry processing and transaction

state management. Integrated callback handling, validation checks, and refund-related workflows. Performed local testing, debugging, and optimization based on review feedback and scenario analysis.

- **Output/result:**
Successfully developed and validated retry-related workflows with improved handling of transaction failures and response management.

Task 4: Local Testing Environment Setup and Regression Testing

- **Description of the task:**
Worked on setting up a local testing environment and exploring regression testing for transaction-related workflows.
- **Approach/method used:**
Understood the existing setup structure and configured dependencies required for end-to-end testing and transaction validation.
- **Implementation details:**
Configured local environment services, performed transaction testing, and started building regression test cases for validating transaction scenarios and workflows. The work included testing multiple cases and analyzing system responses.
- **Output/result:**
Successfully configured the testing environment and initiated regression testing activities for backend workflows.

Task 5: Merchant-Specific Risk Analysis Handling

- **Description of the task:**
Worked on implementing backend handling for merchant-specific risk analysis requirements.
- **Approach/method used:**
Analyzed the existing system and studied the requirement for identifying specific merchant traffic and handling additional response parameters.
- **Implementation details:**
Started implementing identifier-based handling logic and integrated support for additional response fields and refund-related workflows. Initial validation and testing were performed for basic scenarios.
- **Output/result:**
Successfully initiated implementation of merchant-specific handling logic and validated initial workflow behavior.

4.2 System Design / Architecture (if applicable)

Overall System Overview

The backend system followed a service-oriented architecture where multiple backend services handled transaction processing, caching, retry management, callback processing, and testing workflows. APIs were used for communication between modules, while SQL-based databases and Redis caching mechanisms were used for data storage and optimized retrieval.

Data Flow / Workflow Explanation

The transaction workflow involved receiving requests through backend APIs, validating transaction data, processing transaction states, and handling responses through callbacks. Retry logic was implemented for failed transactions, while refund handling was integrated for failure scenarios. Redis caching was used to improve retrieval performance and reduce repeated database operations. Testing workflows included local validation, regression testing, and scenario-based verification.

4.3 Development Process

Methodology Followed

- Agile and Scrum-based development methodology was followed.
- Daily scrum meetings were conducted for task updates and progress tracking.
- Tasks were managed in sprint cycles with regular reviews and demos.

Version Control Practices

- Git was used for version control and code management.
- Merge Requests (MRs) were created for task submissions and reviews.
- Code review feedback was incorporated before merging changes.
- Proper branching and clean commit practices were followed.

Testing and Debugging Process

- Features were tested locally and in staging environments before deployment.
- Debugging was performed using logs, execution tracing, and monitoring tools.
- Edge cases and multiple execution scenarios were validated.
- Sanity testing and production validation were carried out after deployment

4.4 Challenges Faced and Solutions

Technical Issues

- Understanding existing backend transaction workflows and dependencies.
- Handling edge cases in retry and refund scenarios.
- Managing changing requirements and incomplete requirement clarity.
- Configuring local testing environments and validating multiple transaction scenarios.

Team or Communication Challenges

- Coordinating discussions related to changing requirements.
- Understanding workflow dependencies across different backend modules.
- Managing priority changes between ongoing tasks.

How Problems Were Resolved

- Studied existing workflows and documentation before implementation.
- Participated in technical discussions and mentor guidance sessions.
- Performed iterative testing and scenario-based validation.
- Applied modular implementation and debugging techniques for better maintainability.

4.5 Results and Discussion

Final Outcomes of the Work

- Successfully implemented caching-based optimization workflows.
- Improved backend retrieval efficiency and transaction handling processes.
- Developed retry-related transaction handling and validation logic.
- Configured local testing environments and initiated regression testing workflows.
- Started implementation of merchant-specific risk analysis handling.

Improvements Made

- Reduced unnecessary database operations through Redis caching.
- Improved backend response handling and transaction consistency.
- Enhanced reliability of transaction workflows through validation and testing.
- Improved understanding of testing and debugging practices in enterprise systems.

4.6 Learning Outcomes

Technical Skills Gained

- Backend development using PHP, Java, and Spring Boot
- Redis caching and backend optimization techniques
- Transaction retry handling and callback processing
- API testing, debugging, and regression testing
- SQL-based transaction and data management

Soft Skills (Communication and Teamwork)

- Professional communication during discussions and reviews
- Collaboration with mentors and development teams
- Problem-solving and analytical thinking
- Time management and adapting to changing priorities

Industry Exposure

- Exposure to real-world backend transaction systems
- Understanding enterprise software development workflows
- Experience with testing, debugging, and requirement analysis
- Working with scalable and transaction-oriented backend applications

Chapter 5

Summary and Conclusion

The internship completed at PayU provided valuable exposure to real-world backend development workflows, transaction processing systems, caching mechanisms, testing practices, and enterprise software engineering processes. The internship mainly focused on backend development and optimization tasks involving Spring Boot, PHP, Redis, SQL, and API-based workflows.

During the internship, multiple tasks were performed including implementation of a cache monitoring and retrieval API using Redis, optimization of currency conversion storage and retrieval workflows, development of retry handling logic for failed international transactions, transaction status management, callback processing, refund scenario handling, and exploratory regression testing. Additional work involved local testing environment setup and implementation support for merchant-specific risk analysis workflows.

The internship helped in understanding real-world software development lifecycles including requirement analysis, planning, implementation, testing, debugging, optimization, and code review processes. Exposure was also gained in handling edge cases, analyzing backend workflows, validating transaction scenarios, and adapting to changing business requirements.

The work carried out during the internship improved technical knowledge in backend development, caching techniques, API testing, transaction handling, and database management. It also enhanced problem-solving abilities, analytical thinking, debugging skills, communication, and professional collaboration. Concepts learned in engineering subjects such as Database Management Systems, Software Engineering, Object-Oriented Programming, and Web Technologies were practically applied during the internship.

Overall, the internship experience significantly improved understanding of enterprise backend systems and industry-level development practices. It provided practical knowledge of scalable backend architectures, transaction-oriented workflows, and testing methodologies, thereby increasing industry readiness and preparing for future roles in software engineering and backend development.

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